

DD-KOOPMAN: COMPACT SPECTRAL TRANSFER FOR MULTI-STEP CSI PREDICTION

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ABSTRACT

Multi-step channel state information (CSI) prediction must balance forecasting accuracy against inference cost. We propose DD-Koopman, a compact predictor that cleans the observed history, lifts it into the delay–Doppler plane, and evolves it there: its poles and per-mode gains are generated from the observation; the diagonal core is fixed across horizons and satisfies an exact semigroup law. A calibrated finite-window readout with zero-gated corrections completes the model. A closed-form analysis separates input cleaning, spectral propagation and readout calibration. On the official TDD grid it reaches 0.1402 mean NMSE against the 21.9M-parameter reference’s 0.1246, with $126.5\times$ fewer parameters and $4.98\times$ lower latency. Across three seeds it improves mean NMSE on unseen propagation, each paired 95% CI below zero. DD-Koopman is a compact alternative to generic CSI backbones with a structured spectral parameterization.

Index Terms— Channel prediction, CSI, operator learning, delay–Doppler, parameter efficiency

1. INTRODUCTION

The CSI prediction task is to forecast the next four channel states from sixteen historical observations on a $4\times 4\times 2$ MIMO array over 300 subcarriers; adaptive modulation, handover and beam management consume its output. The CSI-4CAST benchmark [1] standardises the splits and the metric and publishes a reference model that reaches a mean normalised mean square error (NMSE) of 0.1246 on its official 162-setting TDD-regular grid, at the cost of 21,913,750 parameters. Our harness reproduces the published number within 0.105% (the benchmark does not seed its RNG), so every comparison below runs on the same protocol.

Prior work treats the delay domain as an input feature for a generic backbone: the delay-domain view enters as features or a side branch, while the multi-step evolution on top remains a generic sequence model.

DD-Koopman instead places the evolution operator itself in the delay–Doppler plane. A specular channel is a finite sum of complex exponentials; when each component has constant Doppler over the observation window, its one-slot evolution is multiplication by a complex exponential, so the exact predictor is diagonal in that plane. The classical zero-training DFT-grid extrapolator cannot express off-grid Doppler, decaying components or the scene-dependent scattering that sets each mode’s radius; DD-Koopman cleans the observation first and learns all three from it.

Our contributions are:

1. **Compact observation-conditioned spectral transfer.** We place the evolution operator in the delay–Doppler plane: a diagonal core K_c whose poles and per-mode readout gains are generated from the CSI history, preceded by a residual denoiser and delay-aware input processing (173,190 parameters).

Held fixed across horizons, the core satisfies an exact local semigroup with non-expansive powers; the classical DFT-grid extrapolator appears as its unit-radius, zero-offset limit, and training starts from a ridge least-squares spectral warm start on the grid.

2. **An analysis that separates the three error sources.** For the diagonal operator we derive a closed-form bound that splits multi-step prediction error into input-cleaning, propagation and readout-calibration terms, with the single-mode phase law as a special case; a representation proposition shows why a grid-pinned retrain can recover the reported accuracy (§4.4).
3. **A validated accuracy–efficiency operating point.** DD-Koopman reaches 0.1402 mean NMSE against the published reference’s 0.1246 while using $126.5\times$ fewer parameters and running $4.98\times$ faster; the width scan holds competitive accuracy from 58k to 327k parameters, while shrunk and parameter-matched alternative architectures do not reach this range. Across three seeds the model improves mean NMSE on a 432-setting unseen-propagation slice, each paired CI entirely below zero (§4.2, §4.3).

2. RELATED WORK

Benchmark and published baselines. CSI-4CAST [1] provides the training grid, the three test splits, the harness, and the published reference rows we compare against, including the language-model-based predictor LLM4CP [2]. Neural channel prediction and feedback predate the benchmark [3, 4, 5]. Cross-paper NMSE is not comparable, so we use the benchmark’s published reference rows and re-implement the Table 1 models through its harness.

Delay-domain representations and conditioned spectral evolution. OTFS [6] made the delay–Doppler plane the modulation domain, exploiting the same sparsity. Prior work uses the delay domain as an input representation or a side branch: ChannelKAN [7] expands dual-domain features via IDFT, MambaCSP [8] embeds a delay-domain representation next to the frequency view, PCGAE-Net [9] crops to the strongest delay taps, and CSI-BERT2 [10] operates on standard representations of the same history. We instead make the plane the space where the operator is diagonal.

Dynamics-based and state-space prediction. The Koopman programme lifts nonlinear dynamics to linear evolution of observables [11, 12, 13], approximated from data by extended dynamic mode decomposition [14], deep embeddings [15] and neural operators [16, 17]. Structured state-space models [18, 19], selective variants [20] and gated recurrences [21], with CSI adaptations [8, 22], learn input-dependent recurrences in latent coordinates. Our operator is diagonal in an explicit delay–Doppler basis, and we measure what one frozen model retains under shift.

Efficient forecasting backbones. PatchTST [23], iTransformer [24], TimeMixer [25], TSMixer [26], Informer [27], N-BEATS [28] and Mamba [20] provide generic forecasting baselines;

we port, retrain and score them under the budgets stated in Table 1, alongside the attention [29], recurrent [30, 31] and linear [32] families they grew from.

3. METHOD

Per antenna, the history is $\mathbf{H} \in \mathbb{C}^{16 \times 300}$ (16 slots, 300 subcarriers); the target is the next four slots; Fig. 1 gives the signal path.

Lift. Two parameter-free transforms map the observation into the working basis: $a = \text{IDFT}_{\text{sub}}(\mathbf{H})$ converts each slot to its delay profile (300 taps), and $\mathbf{D} = \text{DFT}_{\text{time}}(a)$ converts along time, giving the delay–Doppler plane. A specular channel concentrates in a few delay taps and few Doppler modes, so the dynamics live in a sparse, physically derived coordinate system; the lift has no parameters, and all 173,190 parameters belong to the operator, the residual denoiser, and the gated correction paths. Concretely: a layer-normalised log-power context feeds two MLP heads (16→64→32 for the 16 pole offsets and radius shifts, 16→64→128 for the 4×16 complex gain corrections); a three-filter residual denoiser plus the two gated paths carry the remainder.

Diagonal operator with conditioned eigenvalues. For horizon $h \in \{1, \dots, 4\}$ (target slot $15+h$), the base prediction extrapolates each mode exponentially:

$$\hat{a}[h, l] = \sum_m G_h[m](\mathbf{c}) \mathbf{D}[m, l] \zeta_m(\mathbf{c})^h, \quad (1)$$

where m indexes the 16 Doppler bins, l the 300 delay taps, and applying DFT_{sub} to $\hat{a}$ reconstructs the predicted frequency-domain CSI. The poles are $\zeta_m(\mathbf{c}) = r_m(\mathbf{c}) e^{j2\pi(m+\varepsilon_m(\mathbf{c}))/16}$ with total offset $\varepsilon_m(\mathbf{c}) = \delta_m + \Delta_m(\mathbf{c})$: a learned static offset δ_m plus an observation-conditioned shift $\Delta_m(\mathbf{c})$, each in $(-\frac{1}{2}, \frac{1}{2})$, so $\varepsilon_m(\mathbf{c})$ places every pole continuously between the DFT bins. Both the radii $r_m(\mathbf{c}) \in (0, 1)$ and the offsets depend on the observation through a context $\mathbf{c}$ that summarises the observed Doppler power averaged over delay taps; the core $K_c = \text{diag}(\zeta_m(\mathbf{c}))$ is held fixed across the four horizons. The gains $G_h[m](\mathbf{c})$ are complex, indexed by horizon and mode, shared across delay taps, and modulated per observation from the same context; they calibrate the finite-window readout, while the semigroup property belongs to K_c alone.

Proposition 1 (grid limit and stability). If every component’s Doppler lies on the 16-point grid and stays constant over the window, the canonical readout $g_m = \frac{1}{16} e^{j2\pi \cdot 15m/16}$ with unit-modulus poles reproduces the channel exactly at every horizon through (1): the grid predictor is the architecture’s unit-radius, zero-offset limit. The reported model initializes its angular offsets on this grid (δ_m and $\Delta_m(\mathbf{c})$ both start at zero) and then applies the ridge spectral warm start described below. For the learned core, $0 < r_m(\mathbf{c}) < 1$: the semigroup law $K_c^{h+k} = K_c^h K_c^k$ holds exactly and every power is non-expansive, $\|K_c^h\|_2 \leq 1$ — stability, not the exact recovery of the unit-radius grid limit.

Proposition 2 (fixed-grid expressivity). Pin the poles to any nonzero values and disable the gain conditioning: setting $G_{hm} = (PF_N^{-1})_{hm} \zeta_m^{-h}$ makes (1) realise any time map $P \in \mathbb{C}^{H \times N}$ shared across delay taps, so a grid-pinned model with free per-horizon gains spans every shared linear multi-step predictor.

Theorem 1 (error chain). Fix the context and compare the operator’s h -step action $\hat{A}_h = G_h \hat{K}^h \hat{Z}$ against a reference decomposition $A_h^* = G_h^* K_*^h Z_*$ of the same coordinates, where $\hat{Z}$ is the lifted, denoised history the operator is fed, Z_* its ideal counterpart,

and $\|\hat{K}\|_2, \|K_*\|_2 \leq \rho \leq 1$. Then

$$\|\hat{A}_h - A_h^*\|_F \leq \|G_h\|_2 \rho^h e_Z + \|G_h\|_2 h \rho^{h-1} e_K \|Z_*\|_F + e_G \rho^h \|Z_*\|_F, \quad (2)$$

where $e_Z = \|\hat{Z} - Z_*\|_F$, $e_K = \|\hat{K} - K_*\|_2$ and $e_G = \|G_h - G_h^*\|_2$: add and subtract $G_h \hat{K}^h Z_*$ and $G_h K_*^h Z_*$, then expand $\hat{K}^h - K_*^h = \sum_{j=0}^{h-1} \hat{K}^{h-1-j} (\hat{K} - K_*) K_*^j$, which requires no commutativity. The three terms are the three design roles: input cleaning controls e_Z , pole placement controls e_K , and readout calibration controls e_G . The conditioned heads add a context-variation term on top of (2) when the context itself is input-dependent (measured net in §4.4).

Corollary 1 (single-mode phase law). For one mode where $\hat{K} = \zeta$ and $K_* = \tilde{\zeta}$ share the radius r , the propagation difference in (2) is exactly

$$|\zeta^h - \tilde{\zeta}^h|^2 = 4r^{2h} \sin^2\left(\frac{\pi h(\varepsilon - \tilde{\varepsilon})}{16}\right), \quad (3)$$

(substitute $\zeta = r e^{j2\pi(m+\varepsilon)/16}$, $\tilde{\zeta} = r e^{j2\pi(m+\tilde{\varepsilon})/16}$ and use $|e^{jx} - e^{jy}| = 2|\sin \frac{x-y}{2}|$): an h -dependent phase factor weighted by the radius factor r^{2h} , whose leading term for small offset mismatch is $4r^{2h} (\pi h(\varepsilon - \tilde{\varepsilon})/16)^2$. The conditioned poles and gains give a structured parameterization of the finite-window spectral transfer, and the frozen checkpoints make active use of its off-grid degrees of freedom: 89% of trained poles sit more than 0.05 bins off grid, and all 16 modes move across scenes (§4.4).

Gated corrections. A small convolutional *residual denoiser* acts on the input history before the lift, through a zero-initialised output head, and therefore starts as the identity. Two residual paths then re-enter the *same* operator, each contributing a gated difference of two operator evaluations

$$\hat{y} = \text{base} + g_1(\text{op}(\text{mix}(x)) - \text{base}) + g_2(\text{op}(\text{res}(x)) - \text{base}), \quad (4)$$

Here *mix* applies per-tap adaptive reweighting (initialised to one) and a temporal convolution (initialised to zero); *res* is the unexplained part of a learned delay-profile split, a delay-domain analogue of series decomposition [33]. Both paths pass through the operator *op* of (1), and their scalar gates start at zero, so (4) initialises to exactly the base; each antenna is processed independently with shared weights. Gains initialise from a ridge least-squares spectral transfer.

4. EXPERIMENTS

4.1. Setup

The training grid has 27 subsets (3 channel models $\times$ 3 delay spreads $\times$ 3 speeds), 24,300 training and 2,700 validation samples. The test splits are the *regular* split of 162 settings (channel model $\times$ delay spread $\times$ speed $\times$ six signal-to-noise ratios (SNRs)) and the *robustness* split of 486 settings (phase/burst noise and packet drops crossed with the 27 regular combinations). The *generalization* slice has 432 settings over 72 combinations drawn from channel models B and E, with delay spreads 50/200/400 ns and speeds 3–42 m/s, all absent from training; the metric is the benchmark’s mean NMSE over the four horizons. For paired statistics we first average NMSE over the six SNR settings within each channel-model–delay–speed cell, then bootstrap the paired cell differences (5,000 resamples, 95% CI); the intervals describe variation across channel cells for fixed checkpoints. We train with batch 32, Adam [34] at $5 \cdot 10^{-4}$ with a plateau scheduler and early stopping (600 epochs for the reported configuration, 240 for the ablation table), three seeds.

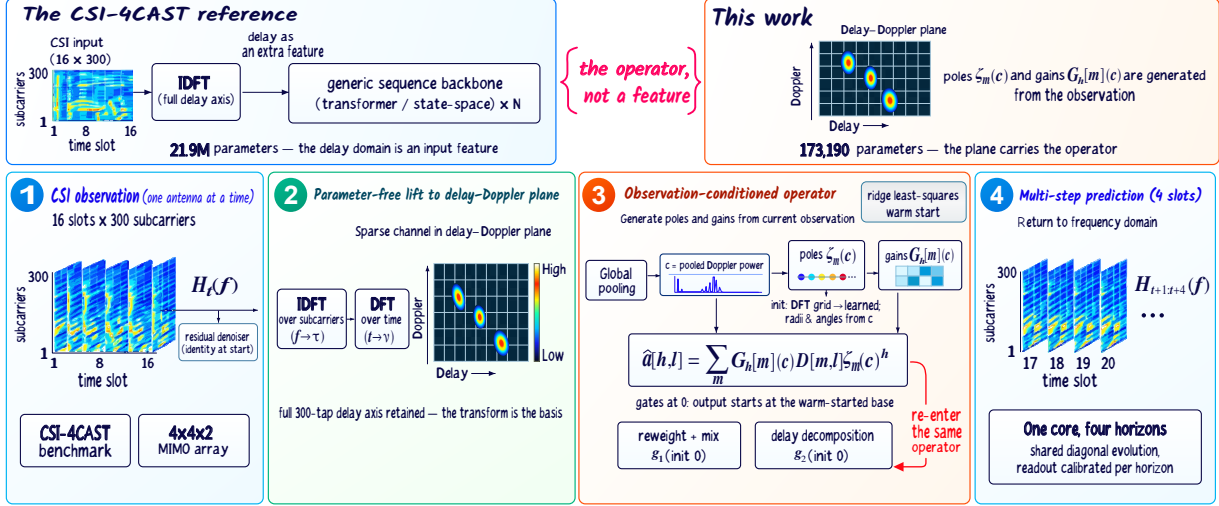


Fig. 1. Overview. The 21.9M-parameter CSI-4CAST reference attaches the delay domain to a generic backbone; this work puts the evolution operator in that plane (173,190 parameters). (1) A residual denoiser, starting as the identity, cleans each antenna’s history. (2) A parameter-free IDFT/DFT lift exposes the sparse delay–Doppler spectrum. (3) A pooled context generates the poles $\zeta_m(c)$ and gains $G_h[m](c)$; two residual paths, gated from zero, re-enter the same operator. (4) The model predicts the next four slots.

Table 1. Mean NMSE on the official 162-setting grid. Published rows (including the classical predictors) come from the benchmark’s reference table; all other rows were trained, or for the DFT-grid extrapolator fitted without training, and scored through the same harness. $\times$ is the ratio to the published row; Budget is the training-epoch cap. MACs per antenna history of size 16×300 : 804.9M (reference), 6.51M (ours), 2857.0M (MambaCSP port), 15.48M (shrunk), 0.48M, 0.77M and 0.92M (transformer rows), 73.77M (TSMixer-L), 0.58M (MLP); the DFT-grid row counts learned MACs only, excluding its analytical FFT cost.

Model	Params	Budget	NMSE	$\times$
CSI-4CAST (published) [1]	21,913,750	—	0.1246	1.00
LLM4CP (published) [2]	—	—	0.1413	1.13
no prediction (published)	—	—	1.8081	14.51
DFT-grid extrap. (zero-train)	0	—	1.3771	11.05
ours, mean of 3 seeds	173,190	600 ep	0.1402	1.13
ours, fixed-grid retrained (3 seeds)	173,190	600 ep	0.1401	1.12
MambaCSP (port, plateau)	881,604	100 ep	0.1468	1.18
CSI-4CAST shrunk (91k)	91,410	240 ep	0.1807	1.45
Transformer (262k)	262,432	240 ep	0.1891	1.52
capacity-matched TF	149,560	240 ep	0.2106	1.69
TSMixer-L [26]	124,337	240 ep	0.2177	1.75
PatchTST [23]	616,560	240 ep	0.2190	1.76
MLP	580,800	240 ep	0.3143	2.52

Re-implemented rows are single seed and early-stopped inside their budgets under the shared validation rule; the fixed-grid retrained row re-trains the reported architecture with its offsets pinned to the DFT grid (three seeds, §4.4); the MambaCSP port stopped at its validation plateau (last improvement at epoch 93 of 100). The remaining ports appear in the text.

4.2. Main comparison and capacity controls

Table 1 and Fig. 2 give the trade-off. Our three-seed mean NMSE is 0.1402 against the published model’s 0.1246 (the seeds score 0.1385, 0.1442 and 0.1378); the per-seed paired differences over the 27 channel cells are +0.0139, +0.0196 and +0.0132, each paired 95% CI entirely above zero.

The benchmark’s classical predictors bound what fixed assumptions deliver: path-delay (0.2300 in distribution) collapses to 0.3231

under shift; autoregressive (0.7874) and Wiener (0.8236) degrade to 1.0469 and 1.0862 against the released checkpoint’s 0.2872; and the fixed 16-point DFT-grid extrapolator the operator replaces reaches only 1.3771, 1.2465 and 1.2790.

We train two compact controls to test whether shrinking the published architecture or a small transformer reproduces this accuracy (Table 1, bottom). Shrinking the published architecture to 91,410 parameters and training it to convergence yields 0.1807, still $1.29\times$ our three-seed mean; a transformer sized to match our parameters reaches 0.2106. The MambaCSP port, trained for 100 epochs to its validation plateau, remains behind the reported model on all three splits (0.1468, 0.2796 and 0.1805) with $5.1\times$ our parameters, so the shift comparison does not rest on an under-trained baseline. The remaining re-implemented ports trail further: GRU 0.3239, TCN 0.8353 and DLinear 0.7941 among the generic families, Mamba 0.3011, iTransformer 0.3877 and TimeMixer 0.5814 among the forecasting ports. Within our own architecture, accuracy is width-insensitive (0.1367–0.1423 across a $5.6\times$, 58k–327k parameter scan), while halving the training windows costs about 17% NMSE: over the tested range, performance is more sensitive to data than to width. Both controls match our parameter scale, not our architecture.

4.3. Behaviour under distribution shift

We evaluate three frozen checkpoints of the reported configuration on the two shifted splits: on the 432-setting generalization slice the three seeds score 0.2685, 0.2767 and 0.2695 against the released checkpoint’s 0.2872. Each paired difference has a 95% CI entirely below zero (−0.0187, −0.0105, −0.0177), and the three-seed mean difference is −0.0156 [−0.0250, −0.0070] over 72 channel cells (5,000-resample bootstrap). On model B the operator leads on all three seeds (three-seed paired difference −0.0400, 95% CI [−0.0531, −0.0271]). On model E the released checkpoint stays ahead (+0.0088 [+0.0046, +0.0132]), though its NMSE is an order of magnitude smaller. The advantage that holds on both unseen models is mobility: below 10 m/s the three-seed difference is

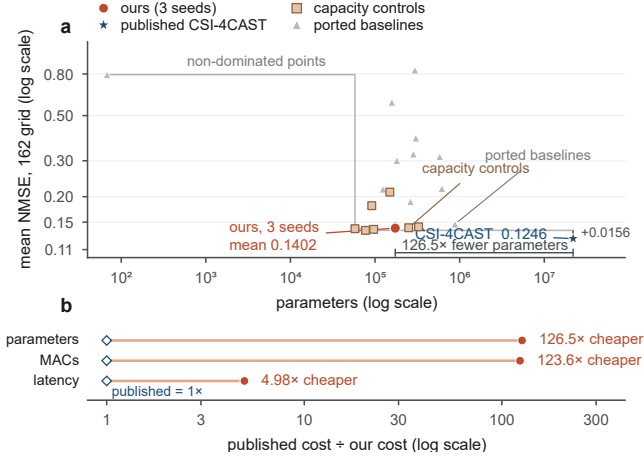


Fig. 2. The accuracy–efficiency plane on the 162-setting grid (log scales): the reported model is the three-seed-validated operating point; the alternative architectures do not reach its accuracy. Star: published model; dots: the three reported seeds; squares: capacity controls (91k shrunk architecture, parameter-matched transformer, 58k–327k capacity scan); triangles: re-implemented baselines. The step line connects the non-dominated points; the 77k square is a single seed, the reported point three-seed-validated. Panel b: the same trade in MACs and latency.

Table 2. Loss-off factorial on the official grid (spectral loss off, 240 epochs, seed 42; ✓ = active, off = removed). Deltas are computed from the tabulated NMSEs.

denoiser	decomp.	poles	NMSE	Δ	$\Delta\%$
✓	✓	✓	0.1397	—	—
✓	✓	off	0.1408	+0.0011	+0.8%
✓	off	✓	0.1425	+0.0028	+2.0%
✓	off	off	0.1458	+0.0061	+4.4%
off	✓	✓	0.1779	+0.0382	+27.3%
off	✓	off	0.1853	+0.0456	+32.6%
off	off	off	0.1862	+0.0465	+33.3%
off	off	✓	0.1968	+0.0571	+40.9%

−0.0204 [−0.0378, −0.0057]; above 30 m/s the two are indistinguishable (+0.0001, CI [−0.0180, +0.0169]).

On the same slice the shrunk architecture scores 0.3039 and the capacity-matched transformer 0.3427, both paired CIs entirely above zero; the plateau-trained MambaCSP port scores 0.2796 (−0.0076, CI [−0.0178, +0.0019]) and the with-loss variant 0.2951, neither significant.

On the benchmark’s 486-setting robustness split the released checkpoint leads consistently: 0.1623 against 0.1772, 0.1735 and 0.1777 across the three seeds, with per-setting paired differences of +0.0112 to +0.0154.

On a second shift axis (channel models A, C and D evaluated outside their training delay-spread and speed ranges; TDD only) the operator stays ahead on the mean, 0.2031 against the published 0.2182 (−0.0151, 95% CI [−0.0256, −0.0054]), with the gain concentrated at 400 ns and heterogeneous across conditions.

4.4. Ablations and the off-grid mechanism

Module factorial. Table 2 runs the full 2^3 loss factorial. The residual denoiser is the dominant single effect (+31.7% average main effect across the four flip pairs), and removing the denoiser together with either other module costs more than the sum of the parts: the

components complement rather than duplicate. One implementation choice is not load-bearing (removing the least-squares warm start costs 0.9%). An auxiliary squared spectral error on the DFT of the four-horizon prediction (a squared variant of FreDF’s frequency-domain supervision [35]) reweights the same regression by Parseval’s identity, yet every matched comparison trains worse with it than without it (two seeds at 600 epochs, one at 240 epochs); we drop the objective.

What the off-grid freedom is worth. A zero-training intervention replaces the frozen model’s total learned offsets $\varepsilon_m(\mathbf{c})$ by constants, every weight unchanged. Setting $\varepsilon_m(\mathbf{c}) = 0$ snaps every pole exactly onto the classical 16-point grid and degrades the official grid from 0.1385 to 0.1923 (+38.8%) with all 162 settings worse, and every constant global offset is worse still: the gain is per-mode, per-sample placement. For $|\varepsilon - \hat{\varepsilon}| < 1$ and $h \leq 4$ the phase factor of Corollary 1 is monotone in h , and the aggregate penalty rises monotonically despite radius attenuation: +23.0/+31.1/+40.5/+45.2% at h1–h4. The offsets must also be resolved finely and jointly (1/8-bin quantisation costs 0.9%, quarter-bin 5.5%, half-bin 12.0%, keeping only the eight largest modes 17.1%), tracking (3). The same intervention degrades all 1,728 settings across the four splits (+17.7% to +38.8%), and a 57,894-parameter variant loses 37.0%; on every shifted slice it pushes the operator below the published model (+0.0288 to +0.0691, all CIs entirely above zero). The frozen predictor’s shift advantage therefore relies on its learned off-grid placement, and the placement matters at every capacity we tested (the intervention costs 37–60% from 58k to 173k parameters). Retraining the same architecture with the offsets pinned to the DFT grid recovers the reported configuration (0.1401 against 0.1402 regular; 0.2688 against 0.2716 unseen, paired CI [−0.0037, −0.0020], three seeds): the calibrated transfer carries the performance; frozen snapping measures how strongly the fitted parameterization leans on the off-grid freedom. On the unseen slice the same factorial attributes the shift advantage to the residual denoiser (+20.9% without it, 95% CI [+0.0431, +0.0713]) and secondarily to the delay decomposition (+4.0%); removing the pole conditioning slightly helps (−0.9%).

4.5. Efficiency

The reported model carries 173,190 parameters and 6,514,464 MACs per antenna sample against the published 21,913,750 and 804,901,376: 126.5× and 123.6× ours. MACs are counted over matmul and convolution only; the parameter-free DFT/IDFT lift and elementwise products are excluded. Forward latency (RTX 4090 D, batch 32, fp32; 3 warm-ups, 12 synchronised repeats, $\pm$ = s.d. over repeats) is 2.24 ± 0.05 ms against 11.15 ± 0.02 ms per scene, a measured scene-level speedup of 4.98×; the MAC count is architectural and does not measure wall-clock time. On the same protocol the MambaCSP port costs 9.88 ms per scene and the capacity-matched transformer 0.40 ms, with NMSE 0.2106 against our 0.1402 seed mean. With MRT beamforming the released checkpoint retains more of the perfect-CSI rate than ours (2.06% vs. 2.70% loss, paired +0.64 pp); system-level rate remains a complementary endpoint.

5. CONCLUSION

DD-Koopman replaces a generic 21.9M-parameter backbone with explicit delay–Doppler evolution: a 173,190-parameter predictor with a diagonal core, ridge-warm-started from a spectral fit on the grid. The error analysis separates input, propagation and readout contributions, while the ablations identify input denoising as the dominant empirical component and reveal the strong sensitivity of the fitted off-grid parameterization. The result: 126.5× fewer parameters, 4.98× lower latency, and the three-seed unseen-propagation advantage.

6. REFERENCES

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